

Lab 4

Title:

To write HTML code to create the given table design

1. Objective:

The objective of this lab work is to design and create a complex table structure using HTML. This lab focuses on the use of table-related attributes such as rows, columns, row spanning, column spanning, and basic styling to represent structured academic data.

2. Tools / Software Used:

- **Operating System:** Windows / Linux / macOS
 - **Text Editor:** VS Code / Notepad++ / Notepad
 - **Web Browser:** Google Chrome / Mozilla Firefox
-

3. Theory:

HTML tables are used to display data in rows and columns. The `<table>` tag is used to create a table, while `<tr>`, `<td>`, and `<th>` are used to define rows, data cells, and header cells respectively.

To create complex table designs, HTML provides attributes like:

- `rowspan` to merge rows
- `colspan` to merge columns
- `border`, `cellspacing`, and `cellpadding` to control table appearance

CSS can be used to improve readability by aligning text and applying font styles. In this lab, internal CSS is used to make headings bold and center-aligned.

4. Algorithm / Procedure:

1. Create a new HTML file.
 2. Define the basic structure of the HTML document.
 3. Add internal CSS for styling table content.
 4. Create a table using the `<table>` tag.
 5. Use `<tr>` and `<td>` tags to define rows and columns.
 6. Apply `rowspan` and `colspan` to merge cells where required.
 7. Insert academic performance data into the table.
 8. Save the file and open it in a web browser to view the table design.
-

5. Source Code:

```
<html>
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
  <title>Table Practice</title>
  <style>
    .bc {
      text-align: center;
      font-weight: bold;
    }

    .b {
      font-weight: bold;
    }
  </style>
</head>

<body>
  <center>
    <h1>Table Practice</h1>
    <table border="1" cellspacing="0"
cellpadding="10" width="50%">
      <tr>
        <td rowspan="3"
class="bc">Department</td>
        <td rowspan="3"
class="bc">Program</td>
        <td colspan="8"
class="bc">Semester-Wise Performance</td>
        <td rowspan="3" class="bc">Grand
Average</td>
      </tr>
      <tr>
        <td colspan="4" class="bc">Odd
Semesters</td>
        <td colspan="4" class="bc">Even
Semesters</td>
      </tr>
      <tr>
        <td class="bc">Sem I</td>
        <td class="bc">Sem II</td>
        <td class="bc">Sem III</td>
        <td class="bc">Sem IV</td>
        <td class="bc">Sem V</td>
        <td class="bc">Sem VI</td>
        <td class="bc">Sem VII</td>
        <td class="bc">Sem VIII</td>
```

```
</tr>
<tr>
  <td rowspan="4">Computer
Science</td>
  <td>Bsc CSIT</td>
  <td>78</td>
  <td>67</td>
  <td>89</td>
  <td>74</td>
  <td>84</td>
  <td>57</td>
  <td>76</td>
  <td>45</td>
  <td>76.75</td>
</tr>
<tr>
  <td>BCA</td>
  <td>84</td>
  <td>78</td>
  <td>74</td>
  <td>57</td>
  <td>67</td>
  <td>89</td>
  <td>45</td>
  <td>76</td>
  <td>72.38</td>
</tr>
<tr>
  <td>Msc CS</td>
  <td colspan="2">78</td>
  <td colspan="2">57</td>
  <td colspan="2">76</td>
  <td colspan="2">45</td>
  <td>76.75</td>
</tr>
<tr>
  <td colspan="9" class="b">Department Average</td>
  <td class="b">76.82</td>
</tr>
<tr>
  <td rowspan="5">Management</td>
  <td>BBA</td>
  <td>78</td>
  <td>67</td>
  <td>89</td>
  <td>74</td>
  <td>84</td>
  <td>57</td>
```

```

        <td>76</td>
        <td>45</td>
        <td>76.75</td>
    </tr>
    <tr>
        <td>MBA</td>
        <td colspan="4">78</td>
        <td colspan="4">82</td>
        <td>76.75</td>
    </tr>
    <tr>
        <td colspan="9" class="b">Department Average</td>
        <td class="b">76.82</td>
    </tr>
    <tr>
        <td colspan="10" class="bc">Overall
        Institutional Average</td>
        <td class="b">76.13</td>
    </tr>
</table>
</center>
</body>

</html>

```

Screenshort:

Table Practice										
Department	Program	Semester-Wise Performance								Grand Average
		Odd Semesters				Even Semesters				
		Sem I	Sem II	Sem III	Sem IV	Sem V	Sem VI	Sem VII	Sem VIII	
Computer Science	Bsc CSIT	78	67	89	74	84	57	76	45	76.75
	BCA	84	78	74	57	67	89	45	76	72.38
	Msc CS	78		57		76		45		76.75
	Department Average									76.82
Management	BBA	78	67	89	74	84	57	76	45	76.75
	MBA	78				82				76.75
	Department Average									76.82
Overall Institutional Average										76.13

6. Output:

The output is a neatly structured table displayed in a web browser showing department-wise and program-wise semester performance, department averages, and overall institutional average using merged rows and columns.

7. Conclusion:

In this lab work, a complex table design was successfully created using HTML table elements. The use of `rowspan` and `colspan` helped in organizing large datasets effectively. This lab improved understanding of HTML tables and their practical application in representing structured academic information.

Print below.....

5. Source Code:

```
<html>
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
  <title>Table Practice</title>
  <style>
    .bc {
      text-align: center;
      font-weight: bold;
    }

    .b {
      font-weight: bold;
    }
  </style>
</head>

<body>
  <center>
    <h1>Table Practice</h1>
    <table border="1" cellspacing="0"
cellpadding="10" width="50%">
      <tr>
        <td rowspan="3"
class="bc">Department</td>
        <td rowspan="3"
class="bc">Program</td>
        <td colspan="8"
class="bc">Semester-Wise Performance</td>
        <td rowspan="3" class="bc">Grand
Average</td>
      </tr>
      <tr>
        <td colspan="4" class="bc">Odd
Semesters</td>
        <td colspan="4" class="bc">Even
Semesters</td>
      </tr>
      <tr>
        <td class="bc">Sem I</td>
        <td class="bc">Sem II</td>
        <td class="bc">Sem III</td>
        <td class="bc">Sem IV</td>
        <td class="bc">Sem V</td>
        <td class="bc">Sem VI</td>
        <td class="bc">Sem VII</td>
        <td class="bc">Sem VIII</td>
```

```
</tr>
<tr>
  <td rowspan="4">Computer
Science</td>
  <td>Bsc CSIT</td>
  <td>78</td>
  <td>67</td>
  <td>89</td>
  <td>74</td>
  <td>84</td>
  <td>57</td>
  <td>76</td>
  <td>45</td>
  <td>76.75</td>
</tr>
<tr>
  <td>BCA</td>
  <td>84</td>
  <td>78</td>
  <td>74</td>
  <td>57</td>
  <td>67</td>
  <td>89</td>
  <td>45</td>
  <td>76</td>
  <td>72.38</td>
</tr>
<tr>
  <td>Msc CS</td>
  <td colspan="2">78</td>
  <td colspan="2">57</td>
  <td colspan="2">76</td>
  <td colspan="2">45</td>
  <td>76.75</td>
</tr>
<tr>
  <td colspan="9">
class="b">Department Average</td>
  <td class="b">76.82</td>
</tr>
<tr>
  <td rowspan="4">Management</td>
  <td>BBA</td>
  <td>78</td>
  <td>67</td>
  <td>89</td>
  <td>74</td>
  <td>84</td>
  <td>57</td>
```

```

<td class="b">76.82</td>
</tr>
<tr>
<td colspan="10" class="bc">Overall
Institutional Average</td>
<td class="b">76.13</td>
</tr>
</table>
</center>
</body>
</html>

```

Table Practice

Table Practice										
Department	Program	Semester-Wise Performance								Grand Average
		Odd Semesters				Even Semesters				
		Sem I	Sem II	Sem III	Sem IV	Sem V	Sem VI	Sem VII	Sem VIII	
Computer Science	Bsc CSIT	78	67	89	74	84	57	76	45	76.75
	BCA	84	78	74	57	67	89	45	76	72.38
	Msc CS	78		57		76		45		76.75
	Department Average									76.82
Management	BBA	78	67	89	74	84	57	76	45	76.75
	MBA	78				82				76.75
	Department Average									76.82
Overall Institutional Average										76.13